Typical Calibration Procedure of Differential Pressure Transmitter

1. Set up the differential pressure transmitter, HART communicator, power supply, hand pump, and the multimeter as below (see below calibration setup file).

2. Make sure the equalizing valve manifold is closed.
3. Apply a pressure to the transmitter equal to a lower range pressure (usually it correspond to 4 mA in the transmitter output). For example we have 0 to 100 mBar calibrated range, then the lower range pressure is 0, or let's say we have -2 psig to 5 psig then we have lower range pressure equal to -2 psig.
4. Read the pressure in the transmitter LCD (or in the HART communicator). Adjust (if any) through the HART communicator so that the output of the transmitter (on LCD) is the same with the applied pressure.
5. Read the mA output of the transmitter by using a multimeter. Adjust (if any) through the HART communicator so that the output of the transmitter (on multimeter) is 4 mA.
6. Apply a pressure to the transmitter equal to an upper range pressure (usually it correspond to 20 mA in the transmitter output).
7. Read the pressure in the transmitter LCD (or in the HART communicator). Adjust (if any) through the HART communicator so that the output of the transmitter (on LCD) is the same with the applied pressure.
8. Read the mA output of the transmitter by using a multimeter. Adjust (if any) through the HART communicator so that the output of the transmitter (on multimeter) is 20 mA.

Typical tools required:

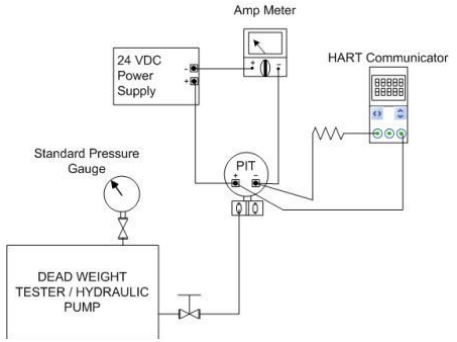
1. 24 VDC power supply
2. Multimeter digital
3. Pneumatic hand pump (up to 600 psig)
4. Hydraulic hand pump (up to 10.000 psig)

5. Low pressure hand pump
6. High precision digital test gauge
7. HART communicator
8. Screwdriver toolkit

Note: point number 1, 2, 7, and 6 of the typical tools above can be replaced by a single multitester available in the market.

Pressure Transmitter Calibration

1. Set up the pressure transmitter as shown in the picture below.



2. Check the pressure transmitter span in the related datasheet.
3. Carry out the five point calibration start from 0% to 100% of span range value.
4. First, set the pressure of a dead weight tester or hydraulic pump or hand pump at pressure equal to 0% of the transmitter span range value.
5. Read the ampere meter reading (wait until the reading is stable) and record it.
6. Repeat step 5 to 6 for 25%, 50%, 75%, and 100% of the transmitter span range value.
7. Repeat step 5 to 6 for downscale start from 100%, 75%, 50%, 25%, and 0%.
8. Calculate the % error for each test point and check whether it still under an acceptable range. (see below)

Formula:

$$\% \text{ Error} = \frac{\text{As Found Value} - \text{Ideal Value}}{\text{Span}}$$

Example:
Range= 4-20mA
Ideal value - Its the value from the range (4, 8, 12, 16 and 20mA)
Span= 20 - 4 = 16mA
As found Value - Its the read out value of the transmitter's output before calibration.

9. If all result still in the acceptable % error range, then the calibration is finish. If there is some test point that shows the result out of acceptable % error range, then do the next step.
10. Set dead weight tester pressure at 0% of the transmitter span range value. The ampere meter reading shall be 4 mA. If the reading is already 4 mA then continue to next step. If the reading not 4 mA then adjust the zero potentiometer of transmitter until the ampere meter read 4 mA. The zero adjustment also available though HART communicator by using zeroing application.
11. Set dead weight tester pressure at 100% of the transmitter span range value. The ampere meter reading shall be 20 mA. If the reading is already 20 mA then back to step 5. If the reading not 20 mA then adjust the span potentiometer of transmitter until the ampere meter read 4 mA. The span adjustment also available though HART communicator by using span application.

Calibration of a Displacer type Level Transmitter.

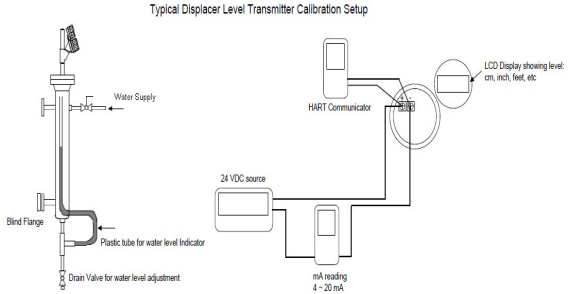
Before we calibrate the displacer level transmitter, we must aware that the transmitter has been set at factory as per process fluid Specific Gravity (SG), while we will use water as the calibration fluid. When the displacer is operated by using actual process fluid, it will show 4 mA at low level (the displacer is not immersed) and 20 mA at high level (the displacer is fully immersed). Thus before we calibrate the transmitter, we need to calculate the equivalence of this buoyancy effect if we use water as the test fluid instead of using actual process fluid (which is in practice are very difficult to get).

Non-Interface Displacer Level Transmitter Calibration Procedure

1. Calculate the equivalence water level
   Zero level = displacer not immersed (no need certain adjustment)
   Calculate the high level using this equation:

(Process Fluid Operating SG / Water SG)*Transmitter Level Range

2. Set up the Displacer Level Transmitter to column or temporary support as shown in the calibration setup file below.



of the lower part flange of the LIT cage as a zero level.

4. By using handheld HART communicator set this level as zero level (see also in the LCD display of the transmitter, it should show zero level).
5. Read the mA output of the transmitter by using a multimeter. Adjust (if any) through the HART communicator so that the output of the transmitter (on multimeter) is 4 mA.
6. Fill the level transmitter chamber with water up to the calculated equivalence high level above.
7. By using handheld HART communicator set this level as high level (see also in the LCD display of the transmitter, it should show high level).
8. Read the mA output of the transmitter by using a multimeter. Adjust (if any) through the HART communicator so that the output of the transmitter (on multimeter) is 20 mA.

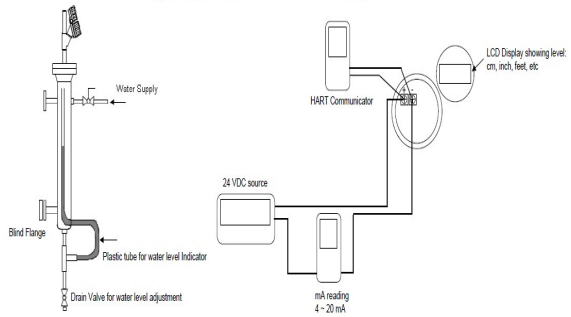
Interface Displacer Level Transmitter Calibration Procedure

1. Calculate the equivalence water level
   Calculate zero level using this equation:

(Lighter Process Fluid Operating SG / Water SG)*Transmitter Level Range
Calculate the high level using this equation:
(Heavier Process Fluid Operating SG / Water SG)*Transmitter Level Range

2. Set up the Displacer Level Transmitter to column or temporary support as shown in the calibration setup file below.

Typical Displacer Level Transmitter Calibration Setup



3. Fill the level transmitter chamber with water up to the calculated equivalence zero level above.
4. By using handheld HART communicator set this level as zero level (see also in the LCD display of the transmitter, it should show zero level).
5. Read the mA output of the transmitter by using a multimeter. Adjust (if any) through the HART communicator so that the output of the transmitter (on multimeter) is 4 mA.
6. Fill the level transmitter chamber with water up to the calculated equivalence high level above.
7. By using handheld HART communicator set this level as high level (see also in the LCD display of the transmitter, it should show high level).
8. Read the mA output of the transmitter by using a multimeter. Adjust (if any) through the HART communicator so that the output of the transmitter (on multimeter) is 20 mA.

Example of equivalence water level calculation:
Level Transmitter Level Range = 38 inch
Water SG = 1
Process Fluid SG = 0.84
Equivalence high level using water = (0.84/1)*38 = 31.92 inch
It means the displacer should show 20 mA while we fill the chamber up to 31.92 inch with water fluid as test fluid.
Interface Application
Level Transmitter Level Range = 38 inch
Water SG = 1
Lighter Process Fluid SG = 0.695
Heavier Process Fluid SG = 0.994

Equivalence zero level using water = (0.695/1)*38 = 26.41 inch
It means the displacer should show 4 mA while we fill the chamber up to 26.41 inch with water fluid as test fluid.
Equivalence high level using water = (0.994/1)*38 = 37.772 inch
It means the displacer should show 20 mA while we fill the chamber up to 37.772 inch with water fluid as test fluid.

Typical tools required:

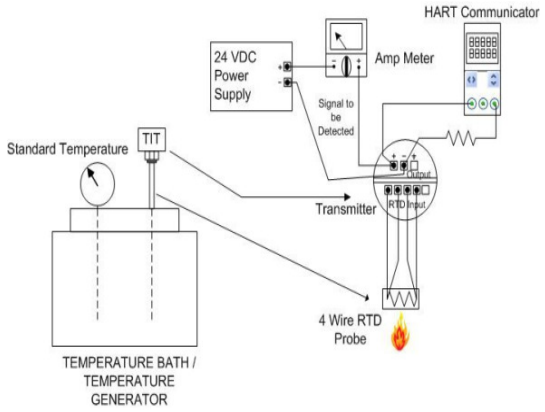
1. 24 VDC power supply
2. Multimeter digital
3. Water Supply Connection
4. HART communicator
5. Screwdriver set
6. Wrench set

Note: This typical maintenance procedure is just an illustration of how to regularly service a displacer level transmitter for academic purpose only. This typical procedure shall not be used as day to day operation guidance. The vendor specific maintenance manual shall be used in detail.

Temperature Transmitter Calibration

Typical Procedure:

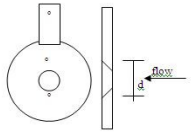
1. Set up the temperature transmitter as shown in the picture below.



2. Switch on the thermo bath / temperature generator power supply.
3. Check the temperature transmitter span in the related datasheet.
4. Carry out the five point calibration start from 0% to 100% of span range value.
5. First, set the temperature at temperature bath to 0% of the transmitter span range value.
6. Read the ampere meter reading (wait until the reading is stable) and record it.
7. Repeat step 5 to 6 for 25%, 50%, 75%, and 100% of the transmitter span range value.
8. Repeat step 5 to 6 for downscale start from 100%, 75%, 50%, 25%, and 0%.
9. Calculate the % error for each test point and check whether it still under an acceptable range.
10. If all result still in the acceptable % error range, then the calibration is finish. If there is some test point that shows the result out of acceptable % error range, then do the next step.
11. Set temperature bath at 0% of the transmitter span range value. The ampere meter reading shall be 4 mA. If the reading is already 4 mA then continue to next step. If the reading not 4 mA then adjust the zero potentiometer of transmitter until the ampere meter read 4 mA. The zero adjustment also available though HART communicator by using zeroing application.
12. Set temperature bath at 100% of the transmitter span range value. The ampere meter reading shall be 20 mA. If the reading is already 20 mA then back to step 5. If the reading not 20 mA then adjust the span potentiometer of transmitter until the ampere meter read 4 mA. The span adjustment also available though HART communicator by using span application.

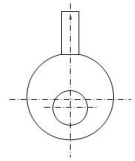
What are the different types of orifice plates and their uses ?

Concentric :
The concentric orifice plate is used for ideal liquid as well as gases and steam service. This orifice plate beta ratio fall between of 0.15 to 0.75 for liquids and 0.20 to 0.70 for gases, and steam. Best results occur between value of 0.4 and 0.6. beta ratio means ratio of the orifice bore to the internal pipe diameters.



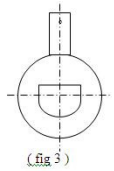
(45º beveled edges are often used to minimize friction resistance to flowing fluid)

Eccentric :
The eccentric orifice plate has a hole eccentric. Use full for measuring containing solids, oil containing water and wet steam. Eccentric plates can use either flange or vena contracta taps, but the tap must be at 180º or 90º to the eccentric opening.



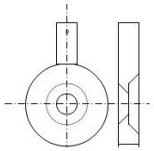
Eccentric orifices have the bore offset from center to minimize problems in services of solids-containing materials.

Segmental :
The segmental orifice place has the hole in the form segment of a circle. This is used for colloidal and slurry flow measurement. For best accuracy, the tap location should be 180º from the center of tangency.



Segmental orifices provide another version of plates useful for solids containing materials.

Quadrant Edge :
It common use in Europe and are particularly useful for pipe sizes less than 2 inchs.



Quadrant edge orifices produce a relatively constant coefficient of discharge for services with low Reynolds numbers in the range from 100,000 down to 5,000.

An operator tells you that flow indication is more ? How would you start checking ?



· *First flushing the transmitter. Flush both the impulse lines. Adjust the zero by equalizing if necessary. If still the indication is more then.*

· *Check low pressure (L.P). side for choke. If that is clean then.*

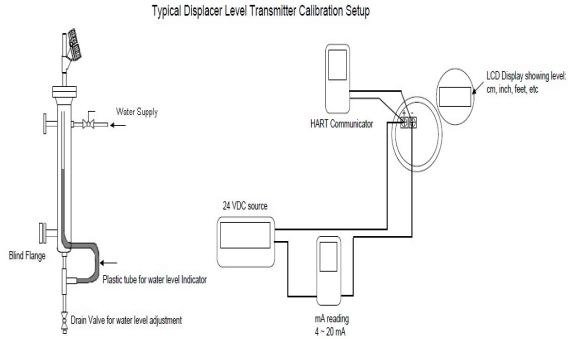
· *Check the leaks on low pressure (L.P.) side. If not,*

· *Calibrate the transmitter.*

DISPLACEMENT TYPE LEVEL TRANSMITTER

Procedure:

1. Determine the operating range and specific gravity of operating fluid from the instrument data sheet.
2. Support the transmitter on a test stand firmly and vertically.
3. Connect water (calibrating fluid) line and clear vinyl hose and other testing equipment as below:



Displacer type Level Transmitter drawing for calibration:

4. Calculate calibration range which converted into water (calibrating liquid) level as follows:
   * Span = Operating range x sp. gr. of operating fluid
   * Zero = Reference level
5. Mark zero and span on the transmitter body and divide into four sections evenly so that they represent 0, 25, 50, 75 and 100% of calibration range.
6. Inject water to the transmitter up to zero % and adjust zero adjustment until 0% of the output obtained.

7. Inject water up to 100% of calibrating range and read the output. The output should be 100% of its range. If it is not, adjust span adjustment.
8. Recheck zero after span adjustment, then repeat step 6 to 8 until desired value obtained.
9. Check output for 0, 25, 50, 75 and 100% of range with level rising and falling. Corresponding mA output are 4mA, 8mA, 12mA, 16mA and 20mA.
10. Every output reading shall be within the limit. If it is not, adjust linearity adjustment and repeat step 6 to 10, every output corresponds with level _____ within acceptable error limit.
11. After the tests have been completed, the instrument shall be identified with colored sticker.
12. Results will be filed and recorded to the applicable calibration form.

Side Mounted vs Top Mounted Level Instrument

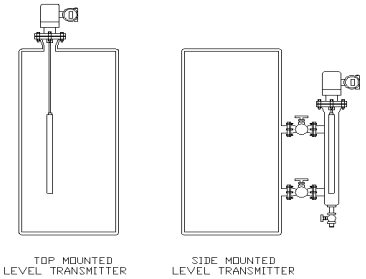
What is the consideration on deciding whether level instrument (transmitter or switch) to be top mounted type or side mounted with external cage type?

Top mounted means the sensor or measuring element of level instrument is inserted into the vessel or tank from the top. Side mounted with external cage means sensor or measuring element is installed integrally in manufacturer prefabricated external cage.

The price of level instrument with external cage is higher than that of top mounted type. This is due to the cost for the cage, moreover for special material such as Duplex Stainless Steel. However, external cage level instrument can have isolation valve installed in between vessel and cage. This arrangement allows instrument maintenance without the need of process shutdown. Therefore external cage level instrument is preferred in most application.

Top mounted level instrument is utilized in some cases such as:

- Buried vessel, since it has no side access.
- Semi-submersible tank such as caisson.
- Liquid which may change properties/form due to temperature change and the use of steam or electrical heat tracing is not feasible.



Above picture shows the arrangement of top mounted level transmitter and side mounted level transmitter. The isolation valve(s) can only be installed on the arrangement of side mounted with external cage type.

DPG3000 Digital Pressure Gauge instructions



All units are factory calibrated prior to shipment.

1) ZERO TRIMMING:

Loosen the set screw retaining the front cover and rotate counterclockwise to remove. You will now have access to the zero adjust potentiometer which is marked with a “Z”. An ideal zero is indicated by a reading of 000 with an intermittently flashing “-” sign. A jeweler screwdriver or an eyeglass screwdriver is a suitable instrument. Be careful NOT to touch the “S” pot to the right, as changing this adjustment will invalidate the factory calibration.

2) BATTERY REPLACEMENT:

After removing the front cover as in 1), loosen the small set screw at the bottom of the Display Assembly and remove the Assembly from the Housing (be careful of the Sensor Leads). The batteries can now be pulled from the holders and replaced with P/N DPG-BAT-C or equivalent.

3) RE-CALIBRATION

This procedure requires a known pressure source of at least +/-0.1% accuracy in order to fully utilize the accuracy potential of the DPG-3000. (If not available, gauge can be returned for re-calibration).

4) PROCEDURE:

A) With 0 psig applied (port vented) adjust zero as per instructions in #1.
B) Apply full-scale pressure to the pressure port and adjust the span (“S”) pot until display reads the correct pressure.
C) Re-check zero and re-adjust the zero (“Z”) pot if required.
D) Repeat steps B) and C) until no further adjustment is required.

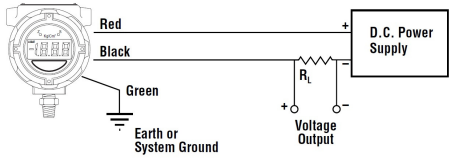
5) 4/20 mA TRANSMITTER OPTION: (–A4 Option)

When equipped with this option the gauge no longer operates from batteries but instead is “Loop Powered”. The loop connection is made to a terminal strip located inside the housing. A voltage of between 9 and 32VDC must be maintained at this connection (Red is positive “+” black is negative “-”) to insure proper operation. Completion of the earth or system ground (Green) is recommended for proper circuit protection. Power supply voltage must be sufficient to maintain a minimum of 9 VDC at the gauge terminals after “dropping” voltage across RL at full scale current (20mA). Example: If RL = 250 ohm then “drop” is 0.02 Amps X 250 ohm = 5 volts. Therefore power supply minimum is 5V + 9V = 14V.

RE-CALIBRATION: Procedure is the same as in **3**,except that there are 2 sets of zero and span adjustments. The front panel controls affect the display and the rear controls (remove electronics as in Step 2) affect the 4/20 mA signal.

NOTE:

DO NOT UNSCREW SENSOR FROM HOUSING, THIS WILL DAMAGE THE UNIT.



Gauge Comparator



This is used as secondary pressure standards, and allow rapid, accurate test of pressure instruments by comparing and instrument on a pressurized test manifold with test gauges of known accuracy and span. The hand operated hydraulic pump has a manual dual volume to rapidly build pressure, and low volume for easy pumping and a gradual approach to the calibration point. The test manifold features a venire for the line adjustment of pump pressure.

Pressure Gauges:

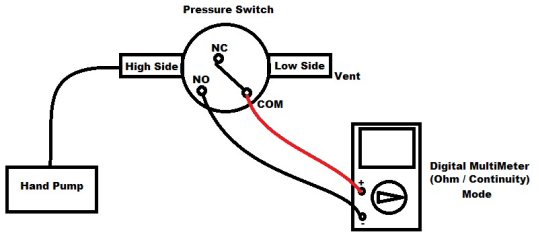
1. Pressure gauges shall be checked by means of hydraulic pressure gauge comparator. This is a means of testing a gauge against a standard pressure gauge.
2. The gauge comparator should be firmly fixed to a bench. A test gauge of range comparable to the gauge under test is fitted to one branch and the hand pump on comparator in order to check the gauge against the readings of the test gauge.

Reading shall be checked for pressures corresponding to 0, 25, 50, 75 and 100% of the range of the gauge under test. Actual gauge reading shall be noted for both rising and falling signals.

3. Test gauges shall have an accuracy of or better than 0 – 25% full scale and shall be periodically checked per specification for accuracy against dead weight tester.
4. Result will be filed and recorded on applicable calibration test/form.

PRESSURE SWITCH:

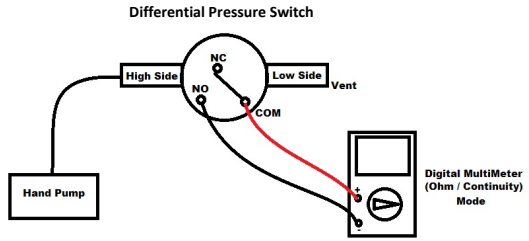
1. Determine the set point and action of switching from instrument date sheet.
2. Hook up testing equipment as below:



3. Check switching action with simulating input signal rising for high pressure service and falling for low. Check dead band.
4. When error found, adjust set point by screw adjustment and repeat until desired value obtained.
5. After the tests have been completed, the instrument shall be identified with a colored sticker.
6. Results will be filed and recorded on applicable calibration form.

DIFFERENTIAL PRESSURE SWITCH:

1. Determine the set point and action of switching from instrument date sheet.
2. Hook up testing equipment as below:



3. Vent low side.
4. Check switching action with simulating input signal rising for high pressure service and falling for low. Check dead band.
5. When error found, adjust set point by screw adjustment and repeat until desired value obtained.
6. After the tests have been completed, the instrument shall be identified with a colored sticker.
7. Results will be filed and recorded on applicable calibration form.

Setpoint:

This is the pressure at which the pressure switch is required to operate. A pressure switch may be set to operate on either a rising pressure (high level alarm) or a falling pressure (low level alarm). Most switches are designed to operate at a 'gauge' pressure setpoint i.e. relative to atmospheric pressure. Some applications require an 'absolute' pressure setpoint i.e. relative to absolute zero pressure, and an absolute pressure switch is required for these. Ideally the range of the switch should be chosen such that the setpoint is between 25% to 75% of this range.

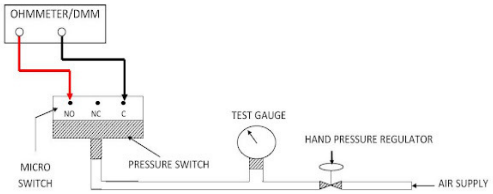
Dead-band or Reset:

This is a setting that determines the amount of pressure change required to re-set the switch to its normal state after it has tripped. The dead-band or reset or switching differential is the difference in the rising and falling pressures at which the pressure switch operates. For a fixed differential output switch this is typically about 1% to 3% of the switch range. For an adjustable differential output switch it may be adjusted from about 5% to 12% of the switch range.

The pressure switch is a ubiquitous device. It is practically everywhere in your plant. But how do you calibrate this simple device? The answer is here. Just follow the simple steps that I have outlined below.
Before you calibrate your pressure switch, confirm the following:

- The setpoint of the pressure switch
- The dead-band of the switch

Also depressurize and isolate the pressure switch from the process. If opening the switch exposes voltages or energy that is not intrinsically safe, please follow the specified procedure for your plant. For example, if in an explosive environment, use a continuously monitoring gas detector to monitor for the presence of explosive gasses.



http://instrumenttoolbox.blogspot.com

Calibration Procedure of the Pressure Switch

Step 1:
Connect the pressure switch to a pressure source e.g air supply via a hand pressure regulator and test gauge, as shown in the diagram above.

Step 2:
Use an Ohmmeter or a Digital Multimeter (DMM) set to the continuity range to check and verify that the switch contacts are as indicated: NO (Normally open) and NC (Normally close).

Step 3:
Connect the Ohmmeter or DMM between the normally open contacts(NO) and the common terminal (C) of the switch. The meter should read "open circuit". Adjust the hand pressure regulator to increase the pressure to the setpoint of the pressure switch until the contacts change over. The meter should now read "short circuit". Note the pressure reading and write it down. This pressure is the switch setpoint for a "rising" pressure.

Step 4:
Increase the pressure to the switch to its maximum rating. Slowly reduce the pressure to the switch until the switch changes over from closed to normally open again. Note and write down this pressure reading. This pressure is the switch setting for a "falling" pressure.

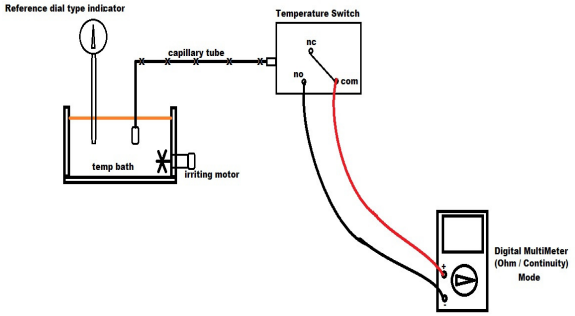
Step 5:
From the readings you have taken work out the pressure difference between the rising and falling pressure settings. This is called the "dead-band" of the switch. The dead-band calculated should be equal to or less than the manufacturers' dead-band.

The maximum dead-band is usually stated by the manufacturer. The switch is unserviceable if the maximum dead-band is more than the manufacturer's recommendation (dead-band on the nameplate of the switch)

To calibrate the switch for a low pressure, go through the steps in this order:
Step 1 to Step 2 to Step 4 to Step 3 to Step 5

TEMPERATURE SWITCH:

1. Determine set point and action of switching from the instrument specification / data sheet.
   Example: 40°C falling (low), 90°C rising (high), etc.
2. Hook up testing equipment as below;

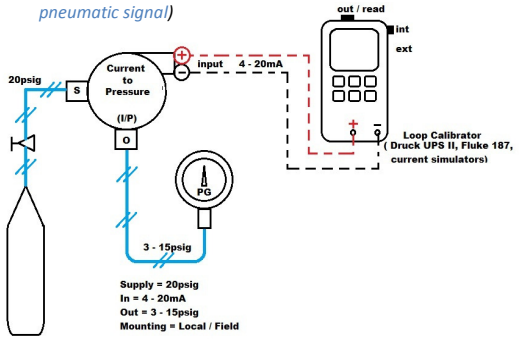


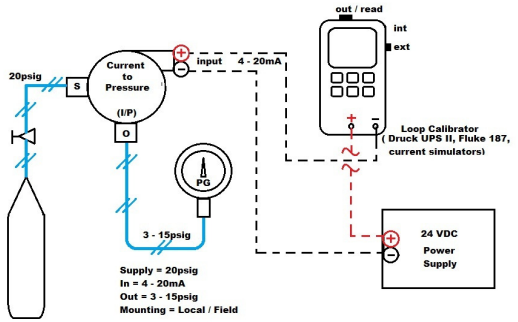
3. Check switching action with simulating temperature rising for high temperature service and falling for low. Check dead band.
4. When error is found, adjust set point by screw adjustment and repeat until desired value obtained.
5. After the tests have been completed, the instrument shall be identified with a colored sticker.
6. Results will be filed and recorded on applicable calibration form.

Note: For ambient temperature as your reference.

CURRENT TO PNUEMATIC (I/P) TRANSDUCER:

1. Hook up first this calibration setup below;
   (note: Broken line is electrical signal and the blue with slash is pneumatic signal)





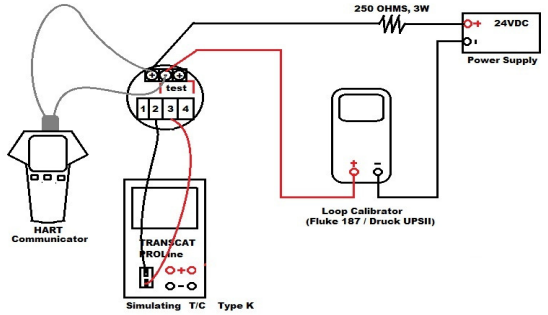
This setup has an external power supply of 24VDC, notice the loop calibrator that the selector switch is at the “ext” position.

2. Inject 0% of input signal (4mA) and check output signal and it must be 3psig. Adjust zero when error found, means not tolerable.
3. Inject 100% of input signal (20mA) and check output signal and it must be 15psig. Adjust span when the output deviates more than the error limit from 100% of its range.
4. Recheck zero after span adjustment and repeat steps 2 to 4 until both are correct.
5. Simulate input signal 0, 25, 50, 75, and 100% (4mA, 8mA, 12mA, 16mA and 20mA) and check output (3psig, 6psig, 9psig, 12psig and 15psig) with input rising and falling. Every output reading shall be within error limit or within tolerance ranges, if not, adjust linearity until desired input to output character obtained.
6. After the tests have been completed, the instrument shall be identified with a colored sticker.
7. Results will be filed and recorded on applicable calibration form.

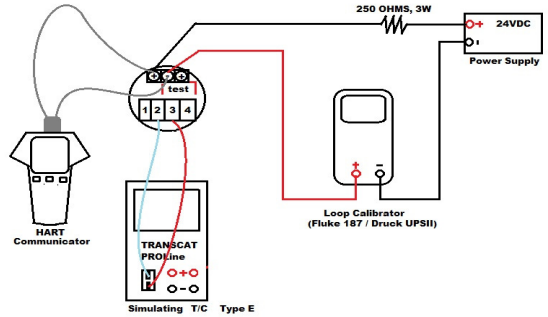
TEMPERATURE TRANSMITTER:

Thermocouple:

A. Type K

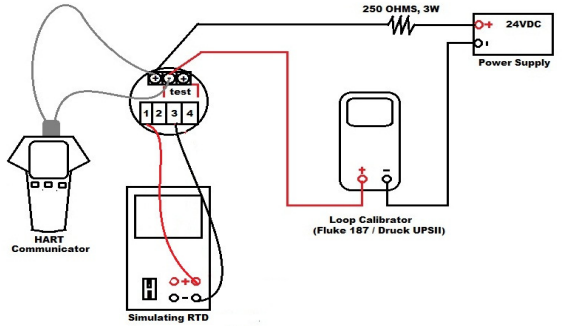


B. Type E



Note: The wires used as an interface between the simulator (TRANSCAT) and Transmitter are different types, notice their colors; Type K has wires black (+) and red (-), while Type E has wires light blue but its white for real (+) and red (-).

Resistant Temperature Detector (RTD):



Note: Also take a good look at the port on where to put those test leads of those different types of temperature sensors.

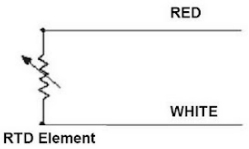
Wiring Arrangement of RTDs:

In order to measure temperature, the RTD element must be connected to some sort of monitoring or control equipment. Since the temperature measurement is based on the element resistance, any other resistance (lead wire resistance, connections, etc.) added to the circuit will result in measurement error. The four basic RTD element wiring methods according to the IEC/ASTM color codes are:

- 2 Wire configuration
- 3 Wire configuration
- 4 Wire configuration
- 2 Wire configurations with compensating loop.

2 Wire configuration RTD:

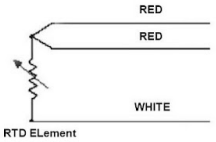
This wire configuration provides one connection to each end of the RTD sensor. This construction is suitable where the resistance of the run of lead wire may be considered as an additive constant in the circuit, and particularly where the changes in lead resistance due to ambient temperature changes can be ignored. This wire configuration is shown below:



Note that the resistance of probe and extension is added to the RTD resistance and will increase the measured value. This could be a source of error in applications where high accuracy is required.

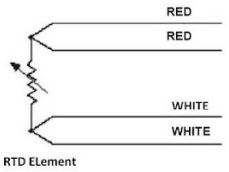
3 Wire Configuration RTD:

This is the standard wire configuration for most RTDs. It provides one connection to one end and two to the other end of the RTD sensor. Connected to an instrument designed to accept three-wire input, compensation is achieved for lead resistance and temperature change in lead resistance. This is the most commonly used configuration.



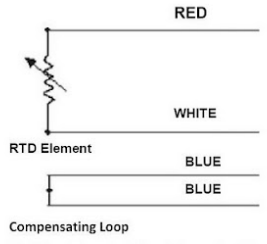
4 Wire Configuration RTD:

This wire configuration provides two connections to each end of the RTD sensor. This construction is used for measurements of the highest precision.



2 Wire Configuration RTD with Compensating Loop:

This is similar to 4 wire configuration RTD except that a separate pair of wires is provided as a loop to provide compensation for lead resistance and ambient temperature changes in lead resistance



VALVE STROKING